EyeSpyJustify: Retinal Disease Classifier & Diagnostic LLM

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Milestone 2

1. Methodology

Our current pipeline involves two key segments. First, we have an ensemble of trained CNNs for multiclass eye disease classification. Second, we conduct a comparison of diagnosis explanation results by prompting a LLM that retrieves from a database built on disease explanations compared to results by prompting MedGemma, which is primarily built on medical data.

For this Milestone, our goal was to build the overarching framework of our ensemble, evaluate the computational and efficiency feasibility of different selected CNNs (both in relation to training and testing), and build a RAG database and test it. This is to ensure that we have a strong architecture foundation before moving on to train and test with our large-scale dataset, fine-tune hyperparameters, and refine prompting and retrieval for diagnosis results.

1.1 CNNs and Ensemble

We selected five pre-trained image classification models to evaluate before building the ensemble: EfficientNet-B0, ResNet18, ShuffleNet-V2, Inception V3, VGG-16.

1.1.1 Image Preprocessing

Specific to this Milestone, we utilized additional image preprocessing to ensure that all fundus images were cropped to eliminate black borders and scaled to the appropriate size required by the image classification models we selected. EfficientNet-B0, ResNet18, ShuffleNet-V2, and VGG-16 use 224x224 pixel images.

1.1.2 Training of Individual, Pre-Trained Models

After testing computational speeds, we determined that Inception V3 and VGG-16 were not computationally viable. We selected EfficientNet-B0, ResNet18, and ShuffleNet-V2 as our final choices. Each model was trained twice, first with 20 epochs for EfficientNet-B0 and 5 epochs for ResNet18 and ShuffleNet-V2, second with 10 epochs for all of the models. For the first run, ResNet18 and ShuffleNet-V2 were trained with batch size of 32 while EfficientNet-B0 was trained with batch size of 128. For the second run, all three models were trained with a batch size of 32. Between the first and second runs, we saw a slight decrease in accuracy between the first and second model for EfficientNet-B0 due to less

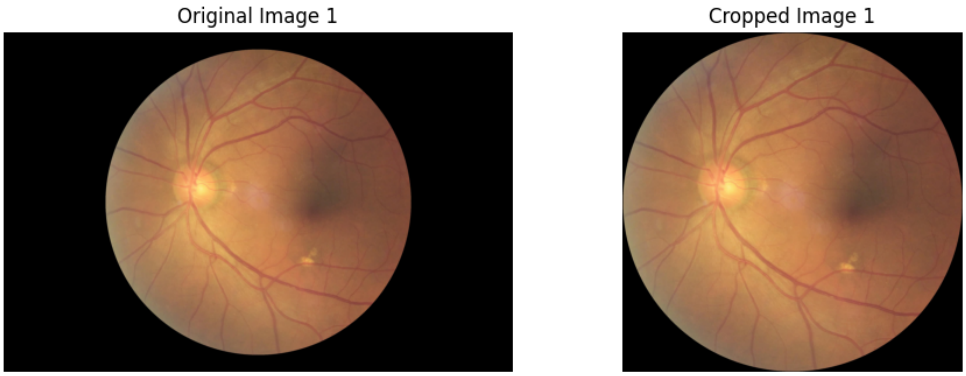


Figure 1: Example of crop

epochs. For ResNet18, we attempted to do multi-label classification and fine tune the learning rate and weight decay however we did not see significant improvements in micro average nor the per-class accuracies between the first and second models. For ShuffleNet-V2, there was slight improvement to accuracy between the first and second model, though still relatively the same as epoch count increased from 5 to 10.

1.1.3 Building of the Ensemble

We built the ensemble by loading the weights of the best first run training models of EfficientNet-B0, ResNet18, and ShuffleNet-V2 and used equal weights to average out the probabilities obtained from each model.

2. Results

2.1 CNNs and Ensemble

We tested each CNN on data from the same dataset that the training data was pulled from. We trained the ensemble on an entirely different dataset since we had already used all the data from our original dataset to train and test the individual CNNs. All the data was preprocessed using the same cropping and resizing technique. No other transformations were performed on the data.

2.2 Individual CNN Results

Figure 2, 3, 4 shows the accuracy for each individual CNN in our ensemble. The testing data comes from the same dataset as training. Each individual model's overall accuracy is around the same: EfficientNet-B0 of 0.61, ResNet18 of 0.54 (micro average), and ShuffleNet-V2 of 0.62. For diseases such as "Cataract", all three models perform relatively well. However, for diseases such as "Diabetic Retinopathy", all three models perform poorly. This is likely caused by class imbalance in the training data, which should be addressed in our full dataset.

	precision	recall	f1-score	support
HYPERTENSION	0.3125	0.2174	0.2564	46
CATARACT	0.7451	0.8539	0.7958	89
AGE_RELATED_MACULAR_DEGENERATION	0.5345	0.4306	0.4769	72
GLAUCOMA	0.5233	0.5625	0.5422	80
MYOPIA	0.6974	0.7571	0.7260	70
DIABETIC_RETINOPATHY	0.5000	0.0588	0.1053	17
NORMAL	0.6469	0.6743	0.6603	872
OTHER_DISEASES	0.5686	0.5539	0.5612	733
accuracy			0.6114	1979
macro avg	0.5660	0.5136	0.5155	1979
weighted avg	0.6060	0.6114	0.6064	1979

Figure 2: Metrics for EfficientNet

	precision	recall	f1-score	support
CATARACT	0.8000	0.7879	0.7939	66
DIABETIC_RETINOPATHY	0.0000	0.0000	0.0000	19
HYPERTENSION	0.0000	0.0000	0.0000	35
MYOPIA	0.7018	0.7407	0.7207	54
AGE_RELATED_MACULAR_DEGENERATION	0.0000	0.0000	0.0000	63
NORMAL	0.5937	0.8003	0.6817	661
GLAUCOMA	0.7500	0.0400	0.0759	75
OTHER_DISEASES	0.7213	0.1615	0.2639	545
micro avg	0.6251	0.4690	0.5359	1518
macro avg	0.4458	0.3163	0.3170	1518
weighted avg	0.6143	0.4690	0.4555	1518
samples avg	0.4737	0.4747	0.4740	1518

Figure 3: Metrics for ResNet

Classification Report:

	precision	recall	f1-score	support
NORMAL	0.6552	0.7213	0.6867	872
DIABETIC_RETINOPATHY	0.0000	0.0000	0.0000	27
GLAUCOMA	0.5821	0.4432	0.5032	88
CATARACT	0.8000	0.8352	0.8172	91
AGE_RELATED_MACULAR_DEGENERATION	0.0000	0.0000	0.0000	73
HYPERTENSION	0.0000	0.0000	0.0000	46
MYOPIA	0.7500	0.7714	0.7606	70
OTHER_DISEASES	0.5484	0.6030	0.5744	733
accuracy			0.6200	2000
macro avg	0.4170	0.4218	0.4178	2000
weighted avg	0.5749	0.6200	0.5959	2000

Figure 4: Metrics for ShuffleNet

2.3 Ensemble Results

Figure 5 and Figure 6 show the metrics and AUROC we obtained for the ensemble. The testing data is from a different dataset as training. The overall accuracy of our ensemble was 0.12. The highest class AUC was 0.53, which was for the "Normal" class. Generally, we observe poor performance across the diseases, likely due to training class imbalance, testing data differences, and equal weighting of the CNNs in the ensemble.

	precision	recall	f1-score	support
NORMAL	1.00	0.12	0.22	49
HYPERTENSION	0.00	0.00	0.00	0
CATARACT	0.00	0.00	0.00	0
AGE_RELATED_MACULAR_DEGENERATION	0.00	0.00	0.00	0
GLAUCOMA	0.00	0.00	0.00	0
MYOPIA	0.00	0.00	0.00	0
DIABETIC_RETINOPATHY	0.00	0.00	0.00	1
OTHER_DISEASES	0.00	0.00	0.00	0
accuracy			0.12	50
macro avg	0.12	0.02	0.03	50
weighted avg	0.98	0.12	0.21	50

Figure 5: Metrics for Ensemble

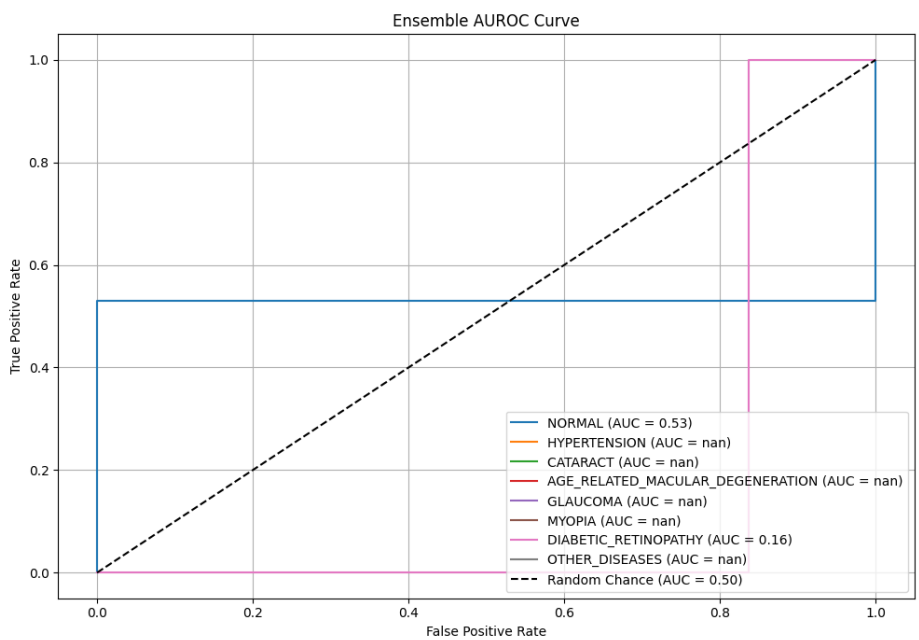


Figure 6: AUROC for Ensemble

3. Analysis

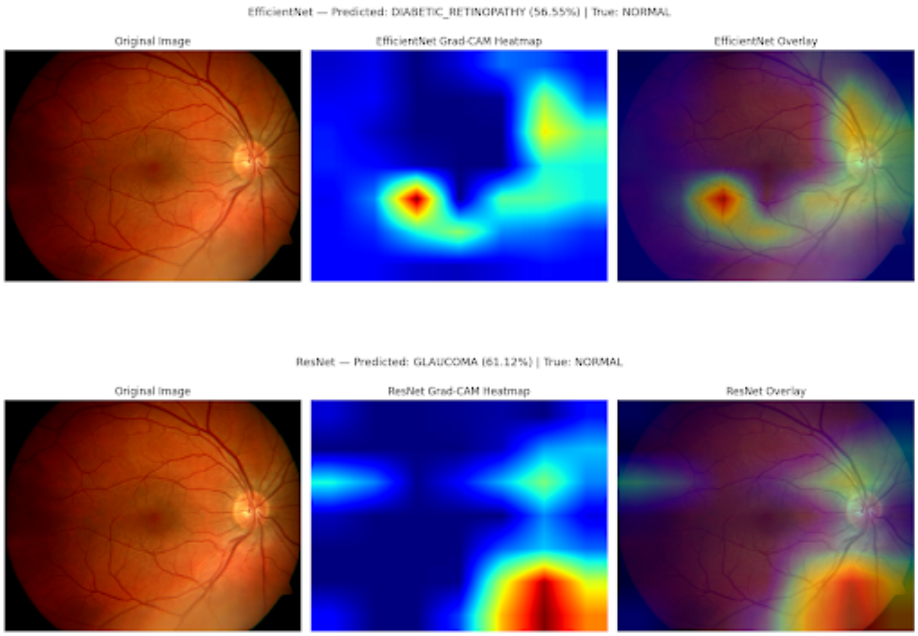
3.1 CNNs and Ensemble

The performance of each individual CNN demonstrated relatively good performance using test sets split from the same dataset as the training set. However, through the ensemble and testing using a test set from a different dataset led to poor performance. Through the evaluation of our current foundational architecture, we observed that this performance may have been caused by a few reasons.

First, the struggle of the CNNs to generalize to images that differ from the quality or style of the original training dataset and overfitting on specific eye diseases. For instance, the EfficientNet model had a higher precision for eyes with a true label of "Age-Related Macular Degeneration" and "Diabetic Retinopathy", while tending to misclassify other labels. In contrast, ShuffleNet had a high precision for eyes with a true label of "Normal", "Cataract", and "Myopia", but had a poor performance for "Diabetic Retinopathy". This is likely due to class imbalance within the training dataset and the need for further image preprocessing of both training and testing datasets.

Second, the equal weights given to each CNN within the ensemble. Adding on to the first point, as each model tends to perform better for different eye diseases, the equal averaging of each model led to misclassifications, even though one CNN correctly classified the test image (as demonstrated in Figure 7 with a test eye image that has the true label of "Normal").

Combined, adding to image preprocessing and finetuning CNN and ensemble hyperparameters should improve performance.



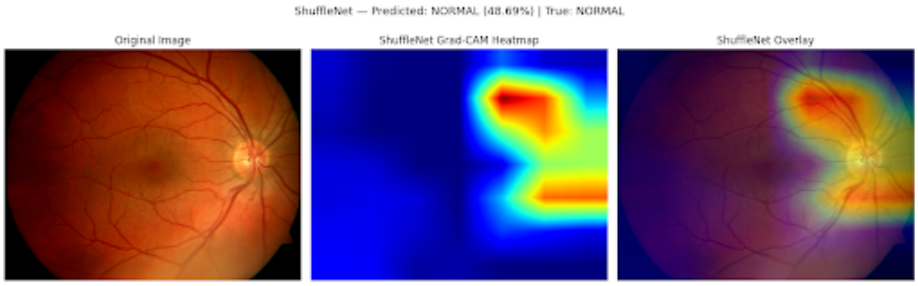


Figure 7: EfficientNet, ResNet, and ShuffleNet heat maps

3.2 LLM and RAG

We believe that our RAG requires more documents in its retrieval database with greater emphasis on disease diagnosis rather than treatment. We plan to increase the number of documents and intentionally choose documents that focus on the diagnosis of the diseases, specifically using fundus images. One other issue we ran into was that for cases where our classification model predicted "Normal" or "Other Diseases", the retrieved documents were completely useless. This is most likely due to the rarity of documents on these classes on the internet. These classes are very general and therefore wouldn't benefit greatly from any external documents. More research is needed in order to overcome this issue.

4. Plan for Additional Analysis

4.1 Image Preprocessing and Training/Testing Dataset

We observed through heatmaps that for some eye diseases such as Glaucoma, the CNNs tended to examine the black background around the eye rather than the optic disc (key indicator identified by doctors). Therefore, for future training and testing, we plan to remove the black background behind the fundus images. Additionally, we plan to try different preprocessing techniques to evaluate their effectiveness. This includes adjusting image coloration and standardizing images that have edges cut off or are stretched. Another important point is that our training dataset included both left and right eyes of the same patient, while our other datasets do not include this. Going forward, we plan to try removing one of the two eyes for the same patient or conduct further augmentation.

4.2 CNN and Ensemble Hyperparameter Tuning

We plan to finetune our current architecture, parameters, and hyperparameters, accounting for how each CNN performs more accurately for different diseases. We will evaluate heatmaps and predictions of each CNN and each disease to identify key parameters and improvements to CNN model architecture and model weights within the ensemble. One concern we had was the class imbalance in our training dataset. Going forward, we will handle the imbalance using techniques like resampling and adjusting class weights in the loss function to penalize minority misclassification higher. Furthermore, for

this Milestone, we mainly focused on single label classification. We hope to account for multilabel classification for all models.

4.3 LLM and RAG

As part of our pipeline, we plan to compare results of prompting our database of eye disease explanations versus prompting MedGemma, which is primarily built on medical diagnosis data. We will adjust MedGemma to fit our needs of explaining the visual and medical differences between eye diseases that were diagnosed, and suggestions for future inspection. We will also look into adding more relevant documents into our retrieval database.

5. Work Plan

Each member was responsible for setting up a pipeline for training the models using an existing image model.Then we utilized the best models to be used in our ensemble. We have also begun developing the front-end application for our project.

GitHub:
`https://github.com/melancholive/Multi-Disease-Retinal-Classification`

Week 1: Adjust and add to image preprocessing steps. Fine-tune hyperparameters and improve model architecture for each individual CNN model.
Week 2: Continuation of improving each CNN's performance. Add more documents to the database for RAG.
Week 3: Fine-tune hyperparameters of ensemble. Adjust MedGemma for retrieval needs.
Week 4: Conduct final adjustments and double-check performance. Connect ensemble and LLM to front-end. Prepare the presentation.